EPIC-I — Research/News Feed — Technical Spec

Status: Draft — for stakeholder review **Date**: 14 July 2026 **Author**: Adrian Hall (Technical Lead), drafted with Claude Code **Scope**: The ninth and final per-epic technical spec. Unlike EPIC-A through EPIC-H, this epic already has substantial design work done — the architecture spec's Section 8 covers source adapters, ingestion, deduplication, classification, and scoring in real detail, validated by the working prototype in `prototypes/research-feed/`. This spec is deliberately lighter: it consolidates what's already decided, adds the member-facing API surface the architecture spec didn't specify, and carries forward the open items already identified there rather than re-deriving them.

Source of truth: the architecture spec's Section 8 (`docs/superpowers/specs/2026-07-14-askapeer-architecture-design.md`) and `docs/AHP_Research_Feed_Design_Conversation.md`/`docs/research-feed-sources-and-roadmap.md`. **Reminder**: EPIC-I is not yet in the PRD's own Section 6.1 MoSCoW list — it was added during architecture design and still needs reflecting back to Paul Gouge and Andrew Renshaw (architecture spec, Section 11 and Section 1).

Contents

---

1. Scope

In scope: source adapters and ingestion (already designed), deduplication and classification (already designed and prototype-validated), the feed API, and — the actual new content in this spec — how a member manages their interest profile, and the relationship between that profile and forum tag-following.

Out of scope: everything the architecture spec's Section 8 already fully specifies is referenced, not repeated, here.

---

2. Data model (as already specified)

Reproduced from the architecture spec, Section 8, for reference only — no changes proposed:

```
research.articles          (id, doi, pmid, other_ids, title,
abstract, journal,
                             published_date, evidence_type,
open_access, source_refs, created_at)
research.ingestion_cursors (source_name, last_run_at, last_cursor)
community.member_interests (handle_id, tag, weight, updated_at)
```

Source adapters (`fetchSince(cursor): RawArticle[]`), deduplication order (DOI → PMID → other IDs → normalised title/year → fuzzy match), rule-based classification, and precomputed scoring (topic match + evidence-type weighting + recency) are all as specified there.

---

3. Member-facing interest management

The architecture spec designs the ingestion and scoring pipeline in detail but doesn't specify how a member actually builds their `member_interests` profile — the working prototype does this via "interest tags on the Profile pane" (per `docs/research-feed-sources-and-roadmap.md`), which this spec formalises as an API:

```
GET  /v1/research-feed/interests
PUT  /v1/research-feed/interests    { tag, weight }     -- upsert;
weight e.g. 0-1 or a small

integer scale, matching whatever
                                                            the
prototype's scoring already

expects — not redefined here
DELETE /v1/research-feed/interests/:tag
```

`member_interests` is handle-scoped (architecture spec, Section 8's data-boundary note) — this is ordinary `community`-schema data, no `identity` involvement, consistent with everything else in that schema.

---

4. `member_interests` vs. EPIC-B's `community.follows`

Worth flagging precisely because it's easy to conflate: `community.member_interests` (this epic, weighted, used to score research-literature relevance) and `community.follows` (EPIC-B, binary, `target_type = tag` — the unified handle/tag follow mechanism resolved 2026-07-14; see EPIC-B's spec, Section 8, and `docs/2026-07-14-technical-specs-open-questions.md`, Section 2) are **two different concepts that happen to both be "a member's interest in a tag."** One is a weighted relevance signal for an external content stream (research articles); the other is a binary follow relationship, now built, for internal content (forum posts, via EPIC-C's personalised feed and EPIC-G's digest). They could plausibly be unified into a single mechanism feeding both the research feed and the forum's personalised feed — or they could legitimately stay separate given they serve different content types with different scoring needs. This spec deliberately doesn't resolve that unilaterally by either building a duplicate mechanism or assuming unification — flagged in Section 9 as still open, now that the tag-follow side of the question is no longer a gap but an actual table to weigh unification against.

---

5. Taxonomy dependency on EPIC-C

EPIC-C's spec (Section 3) already flags that the forum's category/tag taxonomy, Andrew Renshaw's body-area list, and the research feed prototype's `taxonomy.json` are three overlapping-but-not-unified vocabularies. This epic depends on that reconciliation directly: `member_interests.tag` and the classification taxonomy the architecture spec's Section 8 describes should end up being **the same controlled vocabulary** as whatever EPIC-C settles on for forum tags, not a separately-maintained list — otherwise a member's forum tag-following and their research-feed interests would use different, possibly inconsistent vocabularies for what a member experiences as "topics I care about." No new work is proposed here beyond restating this dependency clearly, since EPIC-C's spec already owns resolving it.

---

6. API endpoints

```
GET /v1/research-feed?cursor=...     -- already specified,
architecture spec Section 8:
                                        combines
member_interests weights with precomputed
                                        article scores, returns
an explanation string

GET  /v1/research-feed/interests     -- Section 3, new in this
spec
PUT  /v1/research-feed/interests
DELETE /v1/research-feed/interests/:tag
```

---

7. Carried-forward open items

These are already identified in the architecture spec's Section 11 as EPIC-I pre-launch items — restated here as this epic's responsibility, not re-analysed:

- **DOAJ integration** (predatory-journal legitimacy check) and **Retraction Watch / Crossref integration** (flagging withdrawn papers) — sources identified, integration not yet built.

- **PEDro API access** — worth a direct enquiry to the PEDro team (University of Sydney/NeuRA) given it's the best domain-specific fit found so far, but no confirmed public API.
- **Abstract-redistribution licensing review** — unresolved, relevant if a commercial source (Elsevier/Scopus/Cochrane) is ever added.

---

8. Test plan

- **Interest CRUD**: creating/updating/deleting a `member_interests` row is handle-scoped and has no `identity` involvement (same access-boundary test pattern as every other epic's spec).
- Everything else (deduplication, classification, scoring) is already covered by the working prototype's own validation, per the architecture spec — this spec doesn't add new test surface beyond the member-facing interest endpoints.

---

9. Open questions

- **PRD update**: EPIC-I still needs adding to Section 6.1's MoSCoW list — a standing item from the architecture spec, restated here since it's this epic specifically that's affected.
- `member_interests` **vs.** `community.follows` **unification** (Section 4): a genuine design choice spanning this epic, EPIC-B, EPIC-C, and EPIC-G — the tag-follow mechanism itself is no longer missing (resolved 2026-07-14), but whether it should be unified with this epic's weighted interests remains open.
- **Taxonomy reconciliation** (Section 5): depends entirely on EPIC-C's Section 3 resolution — this epic has no independent path to resolving it.
- **The three carried-forward items** (Section 7): DOAJ/Retraction Watch integration, PEDro enquiry, and licensing review all remain open pre-launch work, unchanged from the architecture spec's own assessment.